Reg. No. | | | | | | | | | | |

[22MSCH1H04]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2023

ANALYTICAL CHEMISTRY

Time: 03 HOURS

Maximum Marks: 50

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. Define determinate and indeterminate errors with their sources.

b. What are the quantitative and qualitative applications of Gas Chromatography?

c. Discuss the principles of HPLC. What are the advantages of HPLC over the LPLC (Low pressure liquid chromatography)?

d. Illustrate and explain the pH titration curve for a weak base *vs.* strong acid titration.

e. How can one differentiate reversible and irreversible electron transfer processes employing cyclic voltammetry?

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Explain the principle, application and advantages of supercritical fluid chromatography.

b. Discuss the various steps involved in the sequencing of a protein.

c. Explain the 'duck shape' nature of a CV plot for a redox couple involving reversible electron transfer process with a suitable example.

SECTION – C

3. Answer the following question. **12**

a. i. What do you mean by standard deviation and Gaussian (normal) distribution?

ii. What are rate theory and plate theory of column chromatography? Explain in detail.

iii. Explain the importance of formation constants of complexes in complexometric titrations.

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